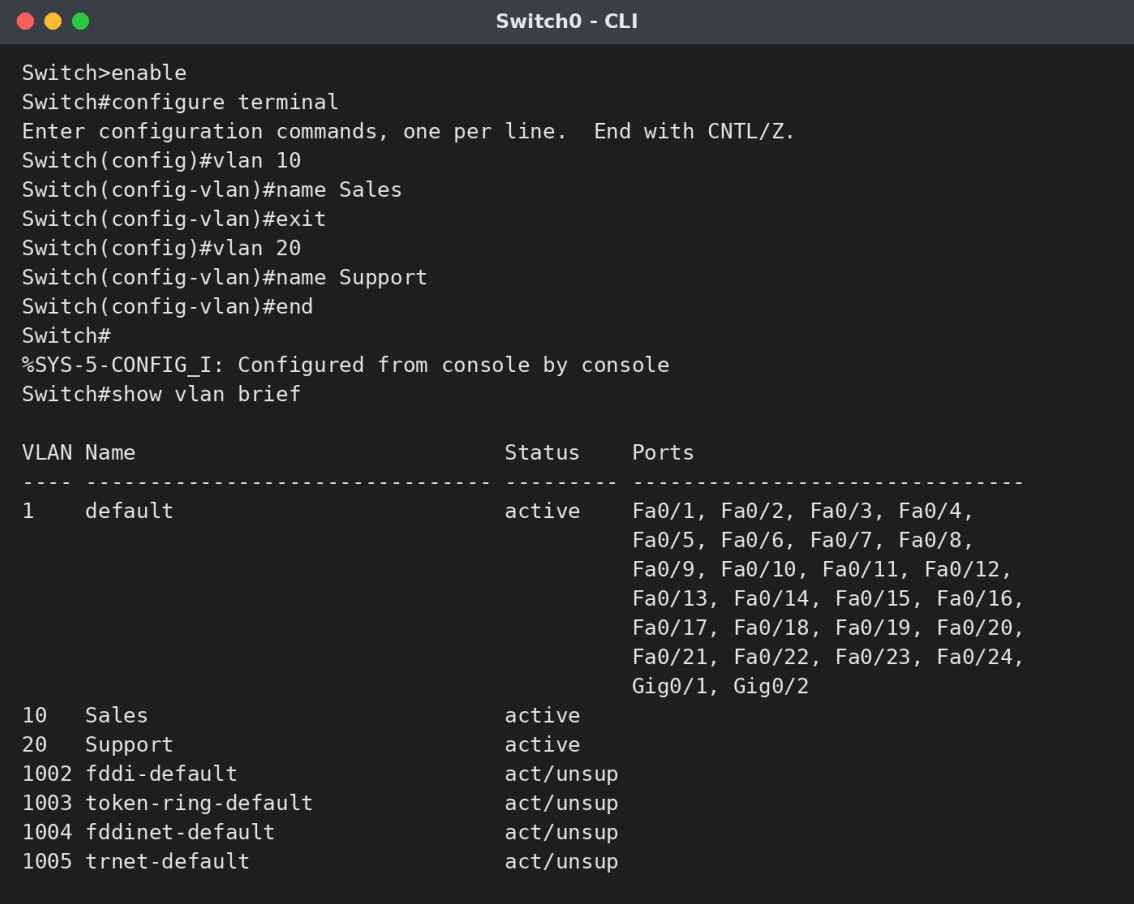


Assignment: Day 8

Switch and Router Configuration Basics

Q1. Log in to Cisco Packet Tracer or any network simulation tool, add a switch, and create two VLANs named 'Sales' and 'Support' with VLAN IDs 10 and 20.

Ans: I opened Cisco Packet Tracer and added a Catalyst 2960 switch (Switch0). In the switch CLI I entered global configuration mode and created VLAN 10 named **Sales** and VLAN 20 named **Support**. The show vlan brief output confirms that both VLANs are active (no ports assigned yet).



```
Switch0 - CLI
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name Sales
Switch(config-vlan)#exit
Switch(config)#vlan 20
Switch(config-vlan)#name Support
Switch(config-vlan)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console
Switch#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/1, Fa0/2, Fa0/3, Fa0/4, Fa0/5, Fa0/6, Fa0/7, Fa0/8, Fa0/9, Fa0/10, Fa0/11, Fa0/12, Fa0/13, Fa0/14, Fa0/15, Fa0/16, Fa0/17, Fa0/18, Fa0/19, Fa0/20, Fa0/21, Fa0/22, Fa0/23, Fa0/24, Gig0/1, Gig0/2
10	Sales	active	
20	Support	active	
1002	fddi-default	act/unsup	
1003	token-ring-default	act/unsup	
1004	fddinet-default	act/unsup	
1005	trnet-default	act/unsup	

Figure 1: Creating VLAN 10 (Sales) and VLAN 20 (Support) on Switch0

Q2. Assign three different switch ports to VLAN 10 (Sales) and two ports to VLAN 20 (Support), then verify the VLAN membership using the show vlan command.

Ans: I used the interface range command to configure several ports at once. Ports Fa0/1 to Fa0/3 were set as access ports in VLAN 10 (Sales) and ports Fa0/4 to Fa0/5 as access ports in VLAN 20 (Support). Five PCs were connected to these ports.

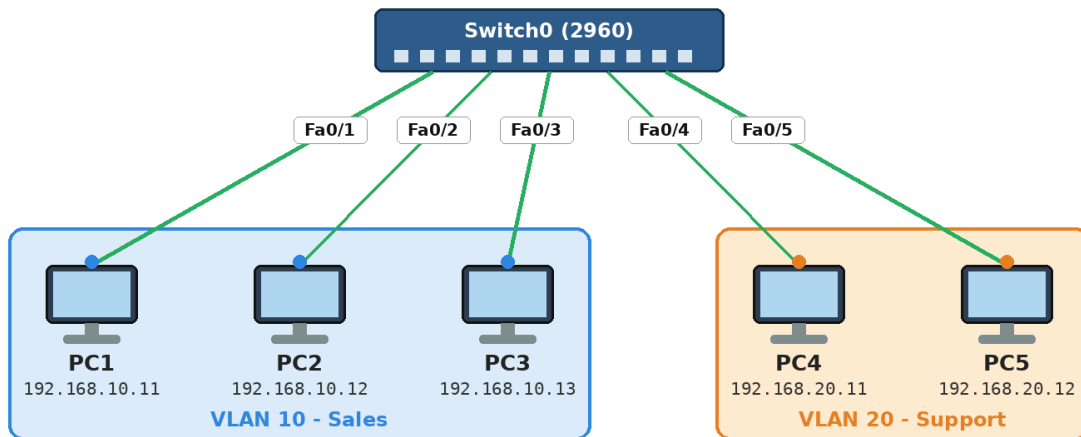


Figure 2: Topology with PCs connected to Switch0 and their VLAN membership

```
Switch0 - CLI

Switch#configure terminal
Switch(config)#interface range fastEthernet 0/1 - 3
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 10
Switch(config-if-range)#exit
Switch(config)#interface range fastEthernet 0/4 - 5
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 20
Switch(config-if-range)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console
Switch#show vlan brief

VLAN Name                Status    Ports
-----
1    default                active    Fa0/6, Fa0/7, Fa0/8, Fa0/9,
Fa0/10, Fa0/11, Fa0/12, Fa0/13,
Fa0/14, Fa0/15, Fa0/16, Fa0/17,
Fa0/18, Fa0/19, Fa0/20, Fa0/21,
Fa0/22, Fa0/23, Fa0/24, Gig0/1,
Gig0/2
10   Sales                  active    Fa0/1, Fa0/2, Fa0/3
20   Support                 active    Fa0/4, Fa0/5
1002 fddi-default          act/unsup
1003 token-ring-default    act/unsup
1004 fddinet-default        act/unsup
1005 trnet-default          act/unsup
```

Figure 3: Port assignment and verification with show vlan brief

The output shows Fa0/1, Fa0/2, Fa0/3 under VLAN 10 (Sales) and Fa0/4, Fa0/5 under VLAN 20 (Support). All remaining ports stay in the default VLAN 1.

Q3. Connect two routers using a serial or Ethernet link in your simulation tool, assign IP addresses to their interfaces, and configure a static route on Router 1 to reach the LAN behind Router 2.

Ans: I connected R1 and R2 with a serial link (R1 is the DCE end, so it has a clock rate). Each router has a LAN behind it with one PC. The IP plan is:

Device	Interface	IP address	Subnet mask	Default gateway
R1	Gi0/0 (LAN 1)	192.168.1.1	255.255.255.0	-
R1	Se0/0/0 (link)	10.0.0.1	255.255.255.252	-
R2	Se0/0/0 (link)	10.0.0.2	255.255.255.252	-
R2	Gi0/0 (LAN 2)	192.168.2.1	255.255.255.0	-
PC1	Fa0	192.168.1.10	255.255.255.0	192.168.1.1
PC2	Fa0	192.168.2.10	255.255.255.0	192.168.2.1

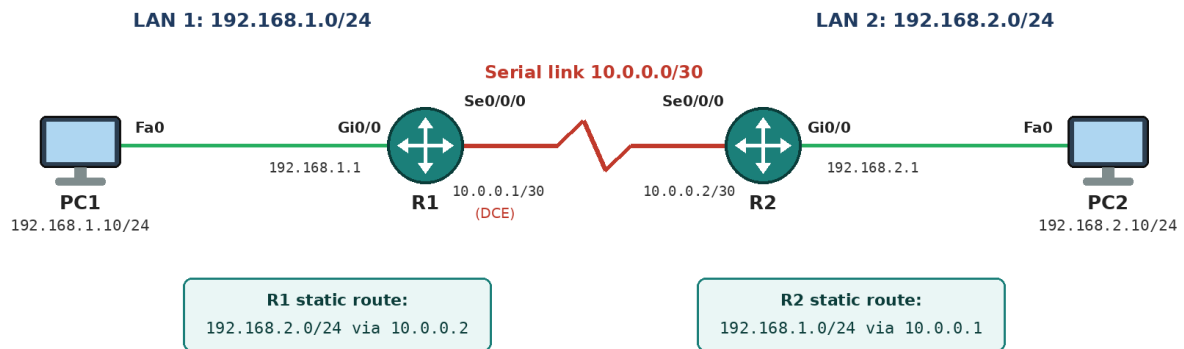
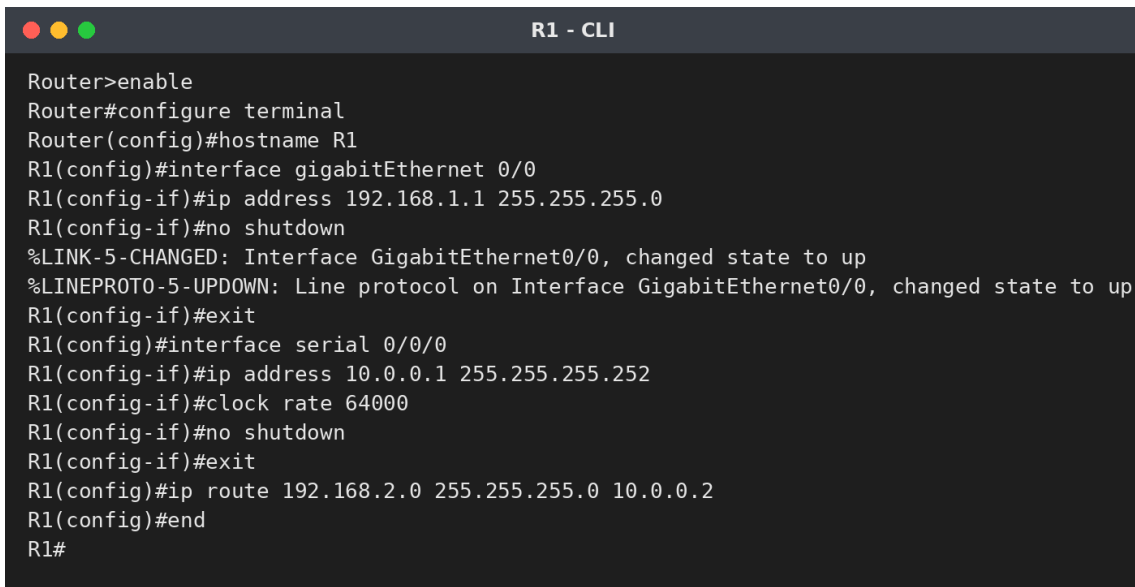


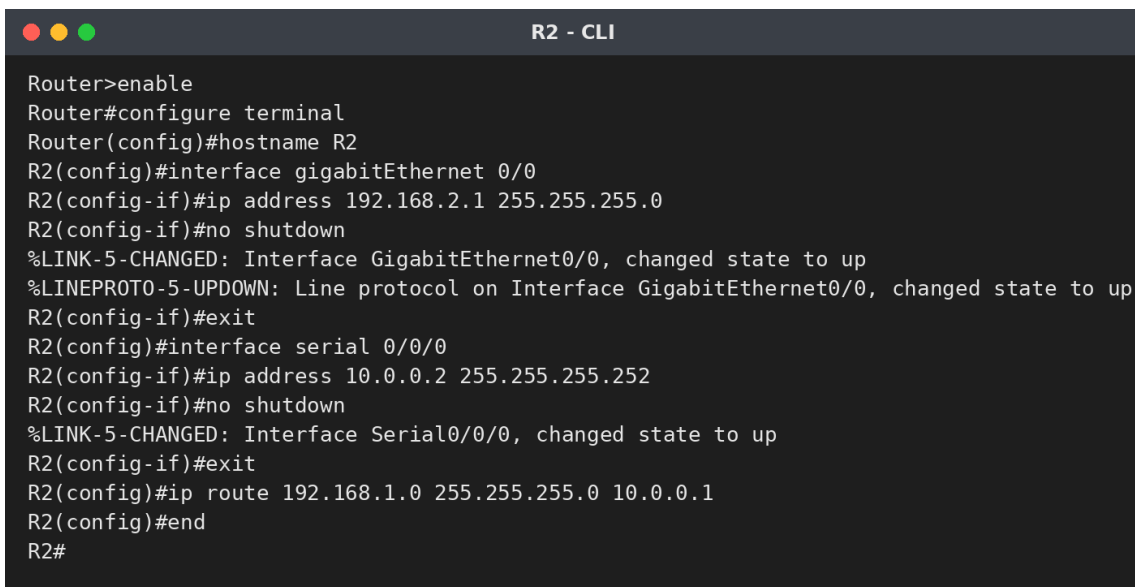
Figure 4: Two routers connected by a serial link with a LAN behind each router

The static route on R1 is `ip route 192.168.2.0 255.255.255.0 10.0.0.2`. It tells R1 that the network 192.168.2.0/24 is reachable through the next hop 10.0.0.2 (R2). I also added a return route on R2, because the reply from PC2 must find its way back to 192.168.1.0/24.

A terminal window titled "R1 - CLI" with a dark background and light gray text. It shows the configuration of router R1, including enabling the terminal, setting the hostname to R1, configuring the GigabitEthernet0/0 interface with IP 192.168.1.1, and the serial 0/0/0 interface with IP 10.0.0.1 and a clock rate of 64000. A static route is added for 192.168.2.0/24 via 10.0.0.2.

```
Router>enable
Router#configure terminal
Router(config)#hostname R1
R1(config)#interface gigabitEthernet 0/0
R1(config-if)#ip address 192.168.1.1 255.255.255.0
R1(config-if)#no shutdown
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
R1(config-if)#exit
R1(config)#interface serial 0/0/0
R1(config-if)#ip address 10.0.0.1 255.255.255.252
R1(config-if)#clock rate 64000
R1(config-if)#no shutdown
R1(config-if)#exit
R1(config)#ip route 192.168.2.0 255.255.255.0 10.0.0.2
R1(config)#end
R1#
```

Figure 5: Interface and static route configuration on R1

A terminal window titled "R2 - CLI" with a dark background and light gray text. It shows the configuration of router R2, including enabling the terminal, setting the hostname to R2, configuring the GigabitEthernet0/0 interface with IP 192.168.2.1, and the serial 0/0/0 interface with IP 10.0.0.2 and a clock rate of 64000. A static route is added for 192.168.1.0/24 via 10.0.0.1.

```
Router>enable
Router#configure terminal
Router(config)#hostname R2
R2(config)#interface gigabitEthernet 0/0
R2(config-if)#ip address 192.168.2.1 255.255.255.0
R2(config-if)#no shutdown
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
R2(config-if)#exit
R2(config)#interface serial 0/0/0
R2(config-if)#ip address 10.0.0.2 255.255.255.252
R2(config-if)#no shutdown
%LINK-5-CHANGED: Interface Serial0/0/0, changed state to up
R2(config-if)#exit
R2(config)#ip route 192.168.1.0 255.255.255.0 10.0.0.1
R2(config)#end
R2#
```

Figure 6: Interface and return route configuration on R2

Verification: the routing table of R1 shows the static route (code S) to 192.168.2.0/24, and PC1 can ping PC2. The first request times out because ARP has to resolve addresses first, which is normal.

```
R1 - CLI

R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/30 is directly connected, Serial0/0/0
L       10.0.0.1/32 is directly connected, Serial0/0/0
    192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, GigabitEthernet0/0
L       192.168.1.1/32 is directly connected, GigabitEthernet0/0
S       192.168.2.0/24 [1/0] via 10.0.0.2
```

Figure 7: show ip route on R1 showing the static route

```
PC1 - Command Prompt

C:\>ping 192.168.2.10

Pinging 192.168.2.10 with 32 bytes of data:

Request timed out.
Reply from 192.168.2.10: bytes=32 time=2ms TTL=126
Reply from 192.168.2.10: bytes=32 time=1ms TTL=126
Reply from 192.168.2.10: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.2.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 2ms, Average = 1ms
```

Figure 8: Ping from PC1 (192.168.1.10) to PC2 (192.168.2.10) is successful

Q4. Simulate a scenario where a device in the 'Sales' VLAN tries to ping a device in the 'Support' VLAN and explain why the ping fails, then suggest what configuration would allow inter-VLAN communication.

Ans: I pinged from PC1 (Sales, 192.168.10.11) to PC4 (Support, 192.168.20.11). The ping failed with "Destination host unreachable".

```
PC1 (Sales) - Command Prompt

C:\>ping 192.168.20.11

Pinging 192.168.20.11 with 32 bytes of data:

Reply from 192.168.10.11: Destination host unreachable.
Reply from 192.168.10.11: Destination host unreachable.
Reply from 192.168.10.11: Destination host unreachable.
Reply from 192.168.10.11: Destination host unreachable.

Ping statistics for 192.168.20.11:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Figure 9: Ping from Sales PC1 to Support PC4 fails

Why the ping fails:

- Each VLAN is a separate Layer 2 broadcast domain. The switch never forwards frames from one VLAN to another, so PC1 and PC4 cannot reach each other at Layer 2.
- The two VLANs use different IP subnets (192.168.10.0/24 and 192.168.20.0/24). PC1 sees that PC4 is in another network and needs a default gateway to send the packet to, but there is no router or Layer 3 device to route between the subnets.

Solution: router-on-a-stick. A router is connected to the switch with one link that is configured as a trunk. On the router, the physical interface is split into sub-interfaces, one per VLAN, each using 802.1Q encapsulation and holding the default gateway address of that VLAN. The PCs use these addresses as their default gateway (192.168.10.1 for Sales and 192.168.20.1 for Support).

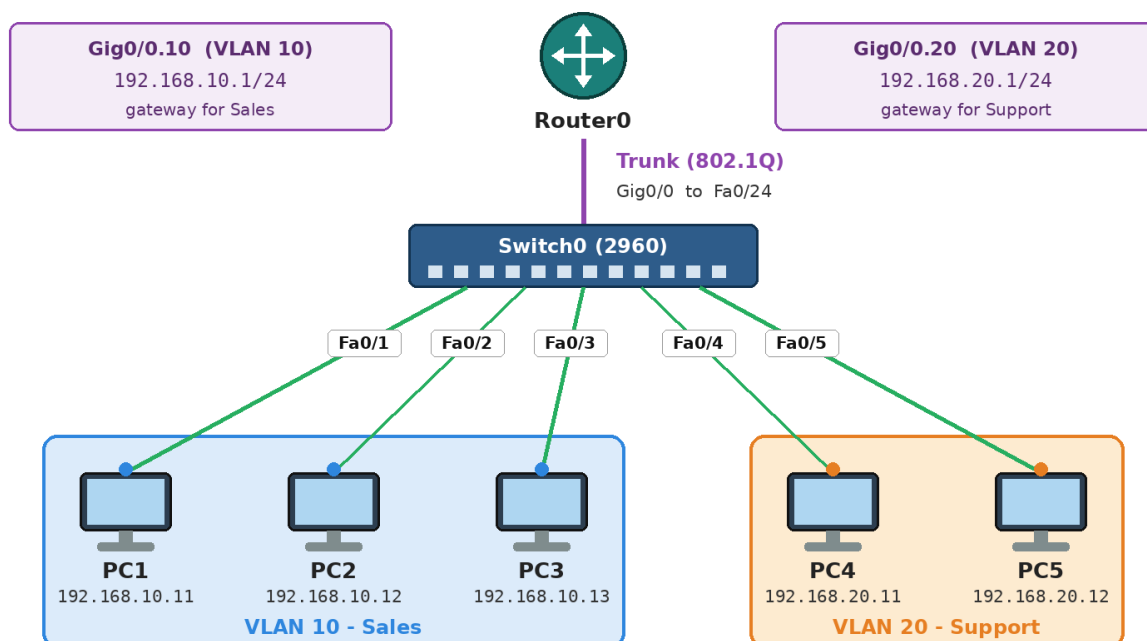


Figure 10: Router-on-a-stick topology for inter-VLAN routing

```
Switch0 and Router0 - CLI

Switch(config)#interface fastEthernet 0/24
Switch(config-if)#switchport mode trunk
Switch(config-if)#end
Switch#

Router(config)#interface gigabitEthernet 0/0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#interface gigabitEthernet 0/0.10
Router(config-subif)#encapsulation dot1Q 10
Router(config-subif)#ip address 192.168.10.1 255.255.255.0
Router(config-subif)#exit
Router(config)#interface gigabitEthernet 0/0.20
Router(config-subif)#encapsulation dot1Q 20
Router(config-subif)#ip address 192.168.20.1 255.255.255.0
Router(config-subif)#end
Router#
```

Figure 11: Trunk port on the switch and sub-interfaces on the router

After this configuration, the ping from PC1 to PC4 works, because the router routes the traffic between VLAN 10 and VLAN 20 (TTL 127 shows one router hop).

```
PC1 (Sales) - Command Prompt

C:\>ping 192.168.20.11

Pinging 192.168.20.11 with 32 bytes of data:

Request timed out.
Reply from 192.168.20.11: bytes=32 time=1ms TTL=127
Reply from 192.168.20.11: bytes=32 time<1ms TTL=127
Reply from 192.168.20.11: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.20.11:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

Figure 12: Ping from Sales PC1 to Support PC4 is successful after inter-VLAN routing

Alternative: A Layer 3 switch can do the same job. Enable routing with `ip routing` and create a switch virtual interface (SVI) for each VLAN, for example `interface vlan 10` with `ip address 192.168.10.1 255.255.255.0` and `interface vlan 20` with `ip address 192.168.20.1 255.255.255.0`.